Social Network Analysis (CS342) | Assignment 10

Introduction to Gephi (Graph Visualization Tool)

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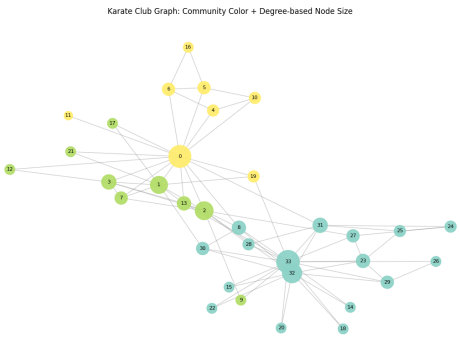
Following tasks need to be performed:

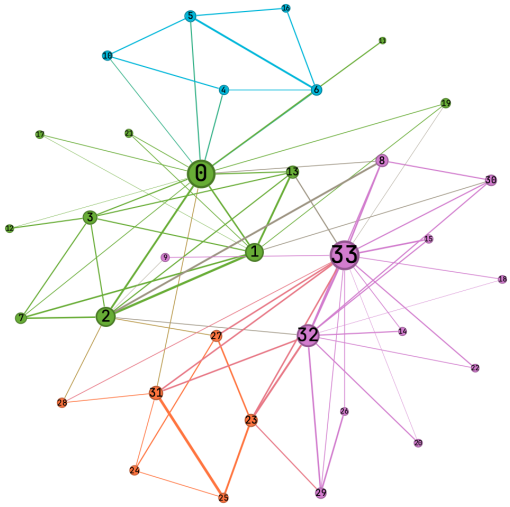
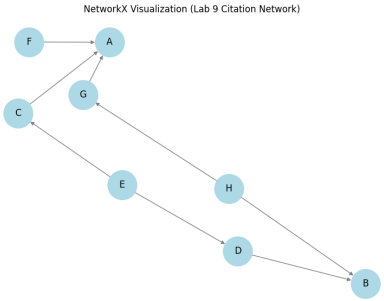
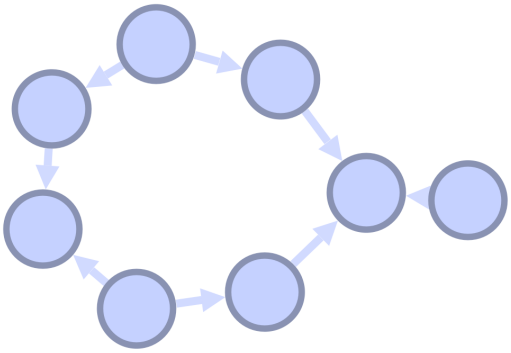
1. Install the "Gephi" software from https://gephi.org/desktop/
2. Explore different features of Gephi (https://gephi.org/quickstart/)
3. Save your netwokx graph in Gephi compatible format (ABCD.gefx file).
4. Open your ABCD.gefx file in the Gephi software.
5. Try different layouts in the Gephi
6. Explore various options for node coloring, sizing, node labeling
7. Explore the statistics tab of the software (degree, diameter, graph density, etc.)
8. Explore the statistics tab of the software for results of HITS, PageRank, connected components
9. Explore community detection tab of Gephi
10. Identify communities in Karate club graph (Appearance→ Nodes→Partition tab in left hand side panel).
11. Use the 'Preview' tab for creating the aesthetic visual representation of your graph.
12. Take any of the graphs created by you in the previous experiments and create a better visual representation for the same using 'Preview' tab.
13. Export your graph in PGF/PNG format.

Deliverables:

1. Graph showing different communities (at least two) based on different node features in the "Karate Club graph".
2. Better visual representation of any of your previous experiment's output. Put both (one generated by NetworkX and one generated by Gephi) the images side by side for comparison.

Karate Club Graphs





Lab 9 Citation Graphs